Ricardo Mendez Vargas

SOFTWARE & DATA ENGINEER | APPLIED AI & AUTOMATION

Chile | ricardo.mendez@gmail.com

https://www.linkedin.com/in/ricardomendez/

PROFILE

Software engineer based in Chile with over 10 years of professional experience, a degree in Automation and Robotics Engineering, and a master's degree in Innovation. Combines hands-on software and data engineering with technical project leadership, AI agent development, process automation, and hardware and software reverse engineering. Delivers practical solutions from requirements and architecture through implementation and production deployment.

CORE EXPERTISE

Software & data: Python, PySpark, Apache Spark, SQL, Snowflake, Snowpark, Hadoop, Django, Streamlit

AI & automation: AI agents, LLM integration, AI-assisted development, workflow automation, web data extraction

Cloud & delivery: AWS, Google Cloud, Docker, Docker Swarm, Portainer, CI/CD; technical project leadership

PROFESSIONAL EXPERIENCE

Software Engineer

Comscore, Inc. | Dec 2022 - Present

- Develop and operate large-scale web data collection pipelines for digital audience measurement, including distributed SERP extraction across multiple international markets.
- Optimize PySpark and Hadoop processing and downstream analytics with Snowflake and Snowpark; build Streamlit applications for self-service data exploration and reporting.
- Develop AI-powered capabilities and autonomous agent workflows, reverse-engineer undocumented legacy software, and deploy production services with Docker, Docker Swarm, Portainer, and CI/CD.

Independent Product Developer | AI-Assisted Development

Self-employed | Jan 2026 - Present

- Designed, developed, tested, and published KiMouse, an Android app for Bluetooth mouse and keyboard control. Managed UX design, closed testing, tester feedback, Google Play publication, and release iterations.
- Use AI-assisted development and rapid prototyping; evaluate local LLMs with Ollama and LM Studio and integrate local and cloud models into development and agentic workflows.

KiMouse on Google Play

IT & IoT Project Developer

INTELIX Chile | May 2017 - Nov 2022

- Developed AgroERP.cl, a web ERP for agricultural businesses using Python and Django, covering requirements, system design, development, implementation, and deployment.
- Built reporting and data capture tools with Node-RED and JavaScript; deployed cloud VPS solutions on Google Cloud and developed IoT architecture and firmware for GPS devices, Arduino, ATmega, and Particle boards.

AgroERP.cl

PROFESSIONAL EXPERIENCE — CONTINUED

Head of Electronics & Innovation

Redd System | Jan 2015 - May 2017

- Led the roadmap for GPS equipment and accessories integrated with web platforms, supporting service continuity, software development, and operations.
- Developed VB.NET interfaces, Java GPS data drivers, Android applications, and Arduino/PIC firmware; analyzed GPS data with SQL and Excel to resolve hardware and software issues.

Electronics Supervisor

Redd System / Tastets System S.A. | Apr 2012 - Dec 2014

- Led electronics engineers and technicians responsible for GPS device testing and software integration.

Electronics Engineer

Redd System / Tastets System S.A. | Jun 2011 - Apr 2012

Electronics Project Specialist | Reverse Engineering

Fábricas y Maestranzas del Ejército (FAMAE) | Jul 2009 - Jun 2011

- Reverse-engineered legacy electronic hardware to reconstruct undocumented circuit functions, interfaces, and signal flows using schematics, measurements, signal tracing, and functional testing.
- Rebuilt technical documentation, diagnosed hardware failures, and developed repair procedures, test fixtures, and diagnostic tools; identified alternatives for obsolete components.

Electronics Technician - CostaDigital

Pontificia Universidad Católica de Valparaíso | Mar 2006 - Jun 2008

EDUCATION & PROFESSIONAL DEVELOPMENT

Master's Degree in Innovation
Pontificia Universidad Católica de Chile | 2014 - 2016

Diploma in Agile Business Development
Universidad del Desarrollo | 2020 - 2021

Diploma in Design and Technology for Smart Cities
Pontificia Universidad Católica de Chile | 2017

Engineering Degree in Automation and Robotics
Universidad Andrés Bello | 2010 - 2013

ELICOS; Certificates II and IV in Renewable Energy
Southbank Institute of Technology, Australia | 2008 - 2009

Electronics Technician
Universidad Técnica Federico Santa María | 2003 - 2006

LANGUAGES

Spanish | English